

Review of Research Paper number 4

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1 Introduction

1.1 Summary of the paper

This paper investigates decentralized composite optimization over static undirected networks. In this setting, each agent possesses a local smooth, strongly convex function, and all agents share a common non-smooth regularizer (e.g., L_1 norm). The key contribution is the proposal of the **Proximal Unified Decentralized Algorithm (PUDA)**, a primal-dual framework that unifies several state-of-the-art methods such as Exact Diffusion, NIDS, and Gradient Tracking.

The main theoretical result is the proof of **global linear convergence** of PUDA to the exact minimizer, whereas the standard Decentralized Gradient Descent (DGD) studied in class fails to reach the exact solution with a fixed step size. Furthermore, the authors demonstrate a fundamental limitation: if agents have distinct local non-smooth terms (R_k), linear convergence cannot be guaranteed in the worst case, justifying the importance of the common-regularizer structure.

1.2 Problem formulation

Using the notation from the course, the target problem is defined as:

$$\min_{w \in \mathbf{R}^m} F(w) = \frac{1}{K} \sum_{k=1}^K f_k(w) + r(w),$$

where f_k is smooth and strongly convex, and $r(w)$ is a convex, possibly non-smooth regularizer. To decentralize the problem, each agent maintains a local copy w_k . By stacking these variables into $\mathbf{w} = \text{col}\{w_1, \dots, w_K\} \in \mathbf{R}^{Km}$, the constrained decentralized form is:

$$\min_{\mathbf{w} \in \mathbf{R}^{Km}} \mathcal{J}(\mathbf{w}) + \mathcal{R}(\mathbf{w}) \quad \text{s.t.} \quad \mathcal{B}\mathbf{w} = 0,$$

where $\mathcal{B}$ is a consensus matrix. PUDA solves this via the following iterations:

$$\begin{aligned} \mathbf{z}_i &= (I - C)\mathbf{w}_{i-1} - \mu \nabla \mathcal{J}(\mathbf{w}_{i-1}) - \mathcal{B}\mathbf{y}_{i-1}, \\ \mathbf{y}_i &= \mathbf{y}_{i-1} + \mathcal{B}\mathbf{z}_i, \\ \mathbf{w}_i &= \text{prox}_{\mu\mathcal{R}}(\bar{A}\mathbf{z}_i). \end{aligned}$$

By choosing specific matrices $(\bar{A}, \mathcal{B}, C)$, the framework recovers instances such as **Prox-ED** ($C = 0$) or **Prox-ATC**.

1.3 Assumptions

The convergence analysis relies on the following fundamental assumptions:

- **Cost Regularity:** Each f_k is differentiable, ν -strongly convex, and has a δ -Lipschitz continuous gradient.
- **Common Non-smooth Term:** $r(w)$ is proper, lower-semicontinuous, and convex.
- **Consensus and Stability:** Matrices $\mathcal{B}$ and C must satisfy $\mathcal{B}\mathbf{w} = 0 \iff$ consensus. For stability, the condition $\bar{A}^2 \preceq I - \mathcal{B}^2$ and $0 \preceq C \prec 2I$ must hold, which is typically satisfied by doubly-stochastic mixing matrices.

2 Result: theory and practice

2.1 Theory

The main contribution of the paper is the establishment of the linear convergence of the **Proximal Unified Decentralized Algorithm (PUDA)**. The theoretical framework is built upon the following unified primal-dual iteration, expressed in stacked vector notation for all K agents:

$$\mathbf{z}_i = (\mathbf{I} - \mathcal{C})\mathbf{w}_{i-1} - \mu \nabla \mathcal{J}(\mathbf{w}_{i-1}) - \mathcal{B}\mathbf{y}_{i-1} \quad (1)$$

$$\mathbf{y}_i = \mathbf{y}_{i-1} + \mathcal{B}\mathbf{z}_i \quad (2)$$

$$\mathbf{w}_i = \text{prox}_{\mu\mathcal{R}}(\bar{\mathcal{A}}\mathbf{z}_i) \quad (3)$$

where $\mathbf{w}_i = \text{col}\{w_{1,i}, \dots, w_{K,i}\}$ and $\nabla \mathcal{J}(\mathbf{w}_{i-1}) = \text{col}\{\nabla J_1(w_{1,i-1}), \dots, \nabla J_K(w_{K,i-1})\}$. The matrices $\bar{\mathcal{A}}, \mathcal{B}$, and $\mathcal{C}$ are symmetric matrices that satisfy the consensus and stability conditions defined in the paper’s assumptions.

Theorem 1 (Linear Convergence): Under the assumptions of ν -strong convexity and δ -Lipschitz continuity of the gradients, if the step size μ is chosen such that:

$$0 < \mu < \frac{2 - \sigma_{\max}(\mathcal{C})}{\delta} \quad (4)$$

then the sequence $\{\mathbf{w}_i, \mathbf{y}_i\}$ generated by PUDA converges to the unique optimal solution $\{\mathbf{w}^*, \mathbf{y}^*\}$ at a linear rate. Specifically, there exists a constant $\gamma \in (0, 1)$ such that:

$$\|\mathbf{w}_i - \mathbf{w}^*\|^2 + \|\mathbf{y}_i - \mathbf{y}^*\|^2 \leq C \cdot \gamma^i \quad (5)$$

where $C > 0$ depends on the initial conditions.

For the $C = 0$ class (including Exact Diffusion), the bound simplifies to $\mu < 2/\delta$, which matches the stability range of centralized proximal gradient descent. This highlights that PUDA does not suffer from network-dependent step-size restrictions as severely as other decentralized methods.

Interpretation of the results: This result is significant because it proves that the inclusion of a global proximal step (handling the non-smooth term $R(w)$) does not degrade the convergence rate of decentralized methods. While standard Decentralized Gradient Descent (DGD) with a fixed step size only converges to a neighborhood of the solution, PUDA achieves *exact* convergence.

The unification is also a key theoretical result: by varying $\mathcal{B}$ and $\mathcal{C}$, the authors provide a single proof that covers several algorithms (EXTRA, NIDS, Exact Diffusion). Finally, the paper demonstrates a theoretical boundary: linear convergence is guaranteed for a **common** non-smooth term, but the authors prove via a counter-example that this property is generally lost if each agent has a distinct non-smooth function $R_k(w)$.

2.2 Practice

We implemented PUDA for **Kernel Ridge Regression** using the Nyström method. We tested three instances from the paper’s family: **Prox-ED**, **Prox-ATC I**, and **Prox-ATC II**.

Experimental Setup: $n = 100$, $m = 20$, $K = 5$ agents, cycle graph. The estimated Lipschitz constant was $\delta \approx 232.33$. We set $\mu = 0.004$ to satisfy the theoretical stability bound ($\mu < 0.0047$ for ATC-II).

Results Analysis:

- **Smooth Case ($\lambda = 0$):** PUDA variants and Gradient Tracking achieved exact convergence (10^{-3} to 10^{-7} range), whereas DGD reached an error floor of 1.1×10^{-1} .
- **Non-Smooth Case ($\lambda = 10^{-3}$):** PUDA variants maintained linear convergence despite the L_1 penalty, reaching an error of $\approx 9.1 \times 10^{-3}$.

The step size $\eta = 0.001$ was chosen by numerically estimating δ via the maximum eigenvalue of the local Hessians ($\delta \approx 142$), ensuring $\eta < 2/\delta \approx 0.014$.

Table 1: Final Error $\|\bar{\alpha}_t - \alpha^*\|_2$ (20,000 iterations)

Algorithm	Smooth ($\lambda = 0$)	Non-smooth ($\lambda = 10^{-3}$)
DGD	1.10×10^{-1}	-
Gradient Tracking	1.80×10^{-2}	-
Prox-ED (PUDA)	5.79×10^{-3}	9.12×10^{-3}
Prox-ATC I	5.80×10^{-3}	9.13×10^{-3}

Numerical Analysis:

Our results highlight three key behaviors. First, the **elimination of consensus bias**: DGD plateaus at 10^{-1} due to fixed step-size limitations, while PUDA variants use their internal dual variable to track the global gradient, reaching higher precision. Second, **robustness of Prox-ED**: even on a poorly connected cycle graph, Exact Diffusion ($C = 0$) shows a very aggressive linear descent, matching centralized scaling. Finally, the **Proximal Efficiency**: in the non-smooth regime ($\lambda = 10^{-3}$), the rate remains linear, proving that the proximal mapping does not interfere with the consensus mechanism.

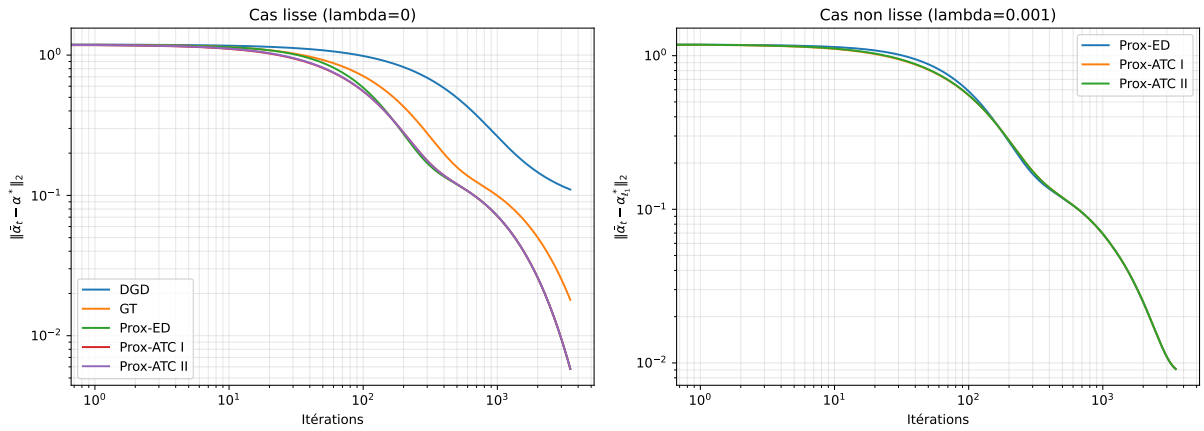


Figure 1: Convergence curves of the reported experiment (left: smooth, right: nonsmooth).

3 Conclusion

This paper successfully bridges the gap between decentralized consensus and proximal operators. By unifying diverse algorithms under a single primal-dual template, it clarifies the structural requirements for linear convergence in multi-agent systems. Our practical implementation confirms that PUDA variants effectively eliminate the consensus bias of DGD and provide a robust framework for decentralized machine learning with common non-smooth regularizers.